YURT 8

A case study for Hut 8

Andres Felipe Pena
andres.p.22455@gmail.com · 347 891 5547
Interactive demo: practicalconstructs.github.io/hut8-casestudy

Yurt 8 is one system that watches every project in the portfolio, generates the next design move on demand, and learns from both, so each pass gets faster, cheaper, and sharper. It is built so leadership sees dollars and days move every single month, and it is built to Hut 8's mission: scaling AI, and using AI to scale itself. The filter for everything in it: keep it simple, cut cost, cut time, raise quality, built to scale.

A pizza tracker for buildings

Domino's earned trust by letting people watch the pizza move. Owners deserve the same view of their capital projects. Every project lives on one chronological tracker, wired to the live data that matters at each stage. Not a status slide. The live state of the work.

Site Selection	Preconstruction	Design	Construction	Procurement	Fit-out	Operations
Land, power, fiber, and tax scored on data before a dollar commits	Budgets, schedules, and contracts lined up as watchable lines	Live model feeds; clashes surface the day they are drawn	RFIs in plain English, with the dollars and days at risk	Long lead equipment tracked like packages, tied to the schedule	Racks, power, cooling; every asset gets its lifetime ID here	The building starts reporting to you, through the same system

Running under every stage: clash detection, BIM, information management, model management, user management, project management, and project delivery. I would also add cost, schedule, safety and QA/QC, commissioning and handover, and energy tracking; those five are flagged for your review.

Review. Generate. Learn. Repeat.

Review: see everything as it happens

- Live feeds of working Revit models, CAD files, and standardized spreadsheets
- RFIs, submittals, and financials translated into plain English for the owner
- Clashes surfaced as they occur, before they become change orders
- ACC is the hub; contractors' Procore, Tableau, Power BI, and Microsoft Project connect by API

Generate: design moves in seconds

- Deterministic guardrails alternate with AI generators, gate after gate, fast and safe
- Test fits and sizing with no third party fees for work that needs no stamp
- Runs in a browser through Autodesk's cloud; nobody needs Revit installed
- Swap cabinet vendors, re-size MEP and containment, re-wire redundant cooling, swap GPUs; every asset number preserved

Learn: every pass sharpens the next

- AWS Bedrock agents on S3 data lakes that grow with every review and generation
- Knowledge gated by review before it teaches anything
- Tied to the codes and standards Hut 8 and its vendors flag as critical
- Grows from stakeholder input and exemplar designs fed in as standardized JSON

> **One shared language.** All three read and write the exact same JSON format `{ coordinates, parameters, relationships, patterns }`. Nothing gets translated between steps, so nothing gets lost between steps.

ACC, kept deliberately simple

Vendors do day to day design in Autodesk Construction Cloud, simplified through clean permissions. For Hut 8 it is a limited tool on purpose: a data source we mine by API. Owners do not need forty modules. They need the tracker.

A folder structure nobody has to hate

```
HUT8-[PROJECT]
├ 01-WIP         each vendor's own working area
├ 02-SHARED      coordinated, clash reviewed
├ 03-PUBLISHED   approved to build, read only
└ 04-ARCHIVE     every version, kept automatically
```

Assets that report for life

ISO 19650, minus the ceremony. Every asset gets one ID at design and keeps it forever: in the model, on the tag, in the sensor feed. IoT site monitoring flows into Operations through the same JSON. The over-engineered version that never survives an American job site is exactly what this is not.

Twenty four months, reported monthly in dollars and days

Build the tool, build the team, then maintain and scale both. Every month closes with one page for leadership: dollars moved, days moved, and which scenario the data came from, alpha, beta, or live. If a month cannot show both numbers, that month failed and I will say so plainly.

Months 1 to 2

Wire into what exists

ACC mining live on one active project. Tracker alpha stands up. Existing contractors and contracts keep their tools; Yurt 8 conforms to the work. **Checkpoint:** first clash caught on screen, priced against the change order it replaced.

Months 3 to 4

Review, fully live

Model feeds, financials, and clash surfacing across the active portfolio. Procore and schedule connections land. **Checkpoint:** avoidable slowdowns identified and priced, monthly.

Months 5 to 6

Generate, first wins

Test fits and sizing on the real cabinet library. Work that took weeks and third party fees now takes seconds. **Checkpoint:** third party design fees eliminated, itemized per run.

Months 7 to 9

Second renaissance engineer

Generate moves into the browser; anyone Hut 8 authorizes can run it without Revit. Hire two arrives. **Checkpoint:** design cycle time, before vs after, in days.

Months 10 to 12

Learn, gated and growing

Knowledge bases go live, gated by review, tied to the codes and standards Hut 8 flags. **Checkpoint:** year one savings tally, line by line.

Months 13 to 18

Portfolio wide, third engineer

Every active project on the tracker. IoT monitoring lights up Operations. New projects start with generative site selection from day one. **Checkpoint:** change order rate vs baseline, portfolio wide.

Months 19 to 24

Full lifecycle, compounding

Site selection through operations on one loop. Every review sharpens generation; every generation feeds the lakes. **Checkpoint:** year two tally, and year three priced from real data.

On meetings: meetings are for decisions. Status lives on the tracker. No decision, no meeting.

On the team: one, then three. Every hire a renaissance engineer who rejects role boundaries. Critical thought, delivered results, first principles, real teamwork. No seat fillers, ever. The system prevents chaos; the team stays drilled for forest fires anyway.

The money

$350K	$4.0M	11x	$10M
Year one ask, all in	Year one savings, conservative	Year one return	Year two savings

Line	Year 1	Year 2
What it costs		
Me, all in	$250K	$350K
Engineers two and three ($260K each)	$0	$520K
Licensing (ACC, APS)	$45K	$90K
Cloud and API tokens	$55K	$90K
Total cost	**$350K**	**$1.05M**
What it returns		
Change orders avoided	$2.4M	$6.0M
Third party design fees eliminated	$0.9M	$1.8M
Schedule value captured	$0.7M	$2.2M
Total savings, conservative case	**$4.0M**	**$10.0M**

Why these savings are real at Hut 8's scale. Hut 8 has 830 MW under construction at its guided $9M to $11M per MW: roughly $8B in flight, matching the $7.5B of project bonds closed in 2026. Studies put design error change orders at 3 to 5% of construction budget; documented BIM clash detection savings reach 10% of contract value. Your Tier 1 partners already run BIM, so these claims take only the owner side residual, well under 1% of watched work. The same benchmarks at base case capture put year one between $8M and $16M; I pitch the conservative number on purpose. And the bonds cost about $1.27M per day in interest at full draw, so a handful of protected days covers this program many times over.

Show my homework. The interactive demo opens the hood: the exact prompt that drove my AI build agents, the architecture, and the sources. I built this in two days with a multi agent setup I designed, and I can build the same without AI: HTML, Python, C#, C, and .NET are home turf. Point me outside my comfort zone and the same financial logic arrives; I learn fast, and I stay motivated and curious.

The 3D in the demo: web native, ships as one file, runs offline, no installs, no servers. When simulation depth earns its cost, the path is NVIDIA Omniverse and the open source OpenUSD ecosystem.

Sources: Hut 8 Q1 2026 results (May 6, 2026); FY2025 Form 10-K; Beacon Point release (July 20, 2026); Vega energization (June 30, 2025); JLL 2025 data center cost index; AIA change order study; Stanford CIFE BIM benchmark; Navigant rework and RFI studies. Full list with links: github.com/PracticalConstructs/hut8-casestudy.